

Chapter 18 – Windows Server Installation, Active Directory Setup & Domain Joining

Windows Server OS Installation in VMware

◆ What is a Server OS?

A **Server Operating System** is designed to provide services to multiple client machines in a network environment.

👉 A normal Windows 10/11 laptop is called a **Client Machine** because it uses services.

👉 A Windows Server machine is called a **Server** because it provides services.

Examples of Server Services

Server Type	Purpose
 Web Server	Hosts websites
 File Server	Stores and shares files
 Mail Server	Handles emails
 DNS Server	Converts domain names into IP addresses
 DHCP Server	Automatically gives IP addresses
 Active Directory Server	Stores users, passwords, computers, policies
 Proxy/VPN Server	Secure remote connectivity
 VMware Server	Handles virtualization

💡 Real-Life Example

Imagine a company with:

- 1000 employees 👤
- 500 laptops 💻
- printers 🖨️
- cameras 📷
- applications 📦

Managing everything manually would be impossible.

So companies use:

- ✓ Windows Server
- ✓ Active Directory
- ✓ Group Policies
- ✓ Domain Controller

for centralized management.



Step-by-Step Windows Server Installation in VMware

◆ Step 1 – Open VMware

Open:

- VMware Workstation / VMware Player

Click:

- ✓ Create a New Virtual Machine

◆ Step 2 – Do NOT Select ISO Initially

When VMware asks for ISO:

- ✗ Do NOT provide ISO file first.

Instead choose:

☒ **I will install the operating system later**

Then click:

 Next

? Why We Do This?

Sometimes Windows Server ISO has:

- installation bugs 
- license agreement page freeze 
- setup detection problems 

If ISO is attached early, installation may hang.

So:

- ☒ First create hardware
- ☒ Then attach ISO later

This is an important troubleshooting technique in virtualization.

◆ Step 3 – Select OS Family

Choose:

Option	Select
--------	--------

Guest OS Family	Microsoft Windows
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Version	Windows Server version
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Example:

- Windows Server 2019
- Windows Server 2022

Then:

 Next

◆ Step 4 – Give VM Name

Example:

- Server01
- DC01
- Company-AD

Then:

 Next

◆ Step 5 – Allocate Disk Space

Provide hard disk space.

Recommended:

- 60 GB or more

Then:

 Next

◆ Step 6 – Customize Hardware

Click:

☒ Customize Hardware

Now configure:

Hardware Recommended

RAM 4 GB

CPU Cores 4 Cores

Then:

- ☒ Close
 - ☒ Finish
-

☒ **VM Created Successfully**

Now:

- ✓ Virtual hardware is ready
- ✓ Machine is created

At this stage:

- no OS installed yet
 - only virtual hardware exists
-

Step 7 – Attach ISO File

Now:

- Right-click VM
- Open Settings

Go to:

☒ CD/DVD

Select:

☒ Use ISO Image File

Browse and select:

 Windows Server ISO path

Click:

☒ OK

Step 8 – Start Virtual Machine

Power on VM.

When screen shows:

Press any key to boot from CD or DVD

 Click inside VM screen

 Press any keyboard key

Now Windows installation starts.

Step 9 – Configure Initial Setup

Select:

Setting	Example
Language	English
Time Format	English US
Keyboard	US

Then:

 Next

Click:

☒ Install Now

Step 10 – Select Windows Edition

Choose:

☒ Windows Server Standard Evaluation (Desktop Experience)

? Why Desktop Experience?

Because it includes:

- ✓ GUI (Graphical Interface)

Without Desktop Experience:

- only command line interface available
- harder for beginners

Step 11 – Create Partition

Click:

- ✓ Custom Install

Create partitions if required.

Then:

- ✓ Select partition
- ✓ Click Next

Installation starts.

If “Getting Files Ready for Installation” Gets Stuck

Possible causes:

Cause	Explanation
 Disk errors	Hard disk corruption
 Bad ISO	Corrupted setup file
 Low disk space	No space to extract files
 Bad sectors	Disk damaged

Troubleshooting Command

Use:

chkdsk

or:

chkdsk /f /r

What CHKDSK Does

Switch Meaning

/f Fix file system errors

/r Recover bad sectors

Important Installation Concept

During installation:

Percentage Process

0–50% Files extracted

After 50% Actual installation starts

Step 12 – Set Administrator Password

After installation completes:

Set strong password:

- ☒ Minimum 8 characters
- ☒ Uppercase letter
- ☒ Lowercase letter
- ☒ Number
- ☒ Symbol

Example:

Admin@123

Server Manager

After login:

Server Manager opens automatically.

◆ **What Can Be Installed from Server Manager?**

Role	Purpose
 DNS Server	Domain name resolution
 DHCP Server	IP distribution
 Web Server	Website hosting
 File Server	File sharing
 Active Directory	User & domain management
 Remote Desktop Services	Remote access

Installing Active Directory Domain Services (AD DS)

◆ **Step 1 – Open Server Manager**

Click:

☒ Add Roles and Features

◆ Step 2 – Installation Type

Choose:

- ☒ Role-based or feature-based installation
-

◆ Step 3 – Server Selection

If multiple servers exist:

- select correct server from pool

Then:

 Next

◆ Step 4 – Select Server Role

Tick:

- ☒ Active Directory Domain Services

Click:

- ☒ Add Features

Then:

 Next → Next → Install

▶ Step 5 – Promote Server to Domain Controller

After installation:

Yellow warning flag appears ⚠

Click:

- ☒ Promote this server to a domain controller
-

Step 6 – Create New Forest

Choose:

☒ Add a new forest

Example domain:

company.local

Then:

 Next

Step 7 – DSRM Password

Enter:

☒ Directory Services Restore Mode Password

Why DSRM Password is Important?

Suppose:

- AD gets corrupted
- Admin deletes AD accidentally
- Server crashes

DSRM password helps:

- ☒ Restore Active Directory
- ☒ Recover Domain Controller

Very important for IT support and disaster recovery.

◆ Step 8 – NetBIOS Name

Server automatically generates:

Example:

COMPANY

Then:

 Next → Next → Install

Server restarts automatically.

What is a Forest?

A Forest is the highest level structure in Active Directory.

Example

Microsoft owns many domains:

- microsoft.com
- xbox.com
- office.com

All related domains together form:

 Forest

OU, Users & GPO

After AD installation:

Go to:

 Tools → Active Directory Users and Computers

From here you can:

- ✓ Create OU
 - ✓ Create users
 - ✓ Manage computers
-

Group Policy Management

Go to:

 **Tools → Group Policy Management**

From here you can:

- ✓ Create GPO
 - ✓ Apply restrictions
 - ✓ Block Control Panel
 - ✓ Set wallpaper
 - ✓ Disable USB
 - ✓ Configure security rules
-

Real Company Scenario

Suppose company blocked:

 Control Panel

using GPO.

Now employee faces:

 No sound issue

But sound settings are inside Control Panel.

What IT Support Does

IT Support engineer logs in using:

- ✓ Admin Account

Because admin account has elevated privileges.

Steps:

- 1 Switch User
- 2 Enter admin username/password
- 3 Open Control Panel
- 4 Fix issue
- 5 Log out

This is why admin access is extremely important.



Domain Joining Practical

To check whether GPO works:

You must join client machine into domain.

Requirements:

- ✓ Windows Server VM
 - ✓ Windows 11 VM
-

◆ Step 1 – Check Network Mode

Ensure both VMs use:

- ✓ NAT Network

or same network type.

◆ Step 2 – Check IP Address

Open CMD in both VMs:

ipconfig

Ensure:

- ✓ Both machines are in same network
-

Example

Machine IP

Server 192.168.1.10

Windows 11 192.168.1.20

Both should belong to same subnet.

◆ Step 3 – Configure Static IP on Server

Open:

Windows + R → ncpa.cpl

Go to:

- Ethernet
- Properties
- IPv4

Set:

Setting	Example
IP Address	192.168.1.10
Subnet Mask	255.255.255.0

Default Gateway Router IP

◆ Step 4 – Configure Windows 11 Network

Same steps:

ncpa.cpl

Go to IPv4 settings.

Set:

Setting	Example
IP Address	192.168.1.20
Subnet Mask	255.255.255.0
Gateway	Same router
DNS Server	SERVER IP

Important

DNS Server must point to:

☒ Windows Server IP

Otherwise domain joining will fail.

Step 5 – Test Connectivity

From Server:

ping 192.168.1.20

From Windows 11:

ping 192.168.1.10

If replies successful:

☒ Network communication working

Step 6 – Join Windows 11 to Domain

Open:

Windows + R → sysdm.cpl

Go to:

- Computer Name

- Change

Select:

☒ Domain

Enter:

company.local

Step 7 – Enter Domain Credentials

Example:

Username	Password
company\hruser1	User password

Click:

☒ OK

System restarts automatically.

◆ **Step 8 – Login Using Domain User**

After restart:

Use:

hr.user1@company.local

Enter password.

First Login Behavior

Windows forces:

☒ Password change

This is called:

Onboarding Process

What Happens After Login?

After successful login:

- ✓ User added to domain
 - ✓ GPO applies automatically
 - ✓ Restrictions enabled
 - ✓ Company policies active
-

How to Verify Domain Join

Open:

sysdm.cpl

Check:

- ✓ Domain name visible

Example:

Member of Domain: company.local

Quick Troubleshooting Tips for IT Support

Problem	Quick Solution
Domain join failed	Check DNS
Cannot ping server	Check firewall/network
GPO not applying	Run gpupdate /force
Login failed	Verify username format
VM not connecting	Check NAT/Bridge mode
AD not opening	Check services
Installation stuck	Run chkdsk
Slow server	Increase RAM/CPU



Important Interview Questions

? What is Active Directory?

A Microsoft directory service used to manage users, computers, policies, and resources centrally.

? What is a Domain Controller?

A server where Active Directory is installed.

? What is a Forest?

Collection of multiple domains and subdomains.

? Why DNS is important in AD?

Because clients use DNS to locate the Domain Controller.

? Why use Group Policy?

To apply security settings and restrictions automatically.

Final Conclusion

This lecture covered complete practical concepts of:

- ✓ Windows Server Installation
- ✓ VMware Configuration
- ✓ Active Directory Installation
- ✓ Domain Controller Setup
- ✓ OU & User Management
- ✓ GPO Application
- ✓ Domain Joining
- ✓ DNS Configuration
- ✓ IT Support Troubleshooting

These concepts are extremely important for:

-  IT Support
-  Cyber Security
-  System Administration
-  Network Administration
-  SOC Operations
-  Windows Infrastructure Management

Chapter 18 – Windows Server Installation, Active Directory Setup & Domain Joining

Q & A

SECTION 1 – Windows Server Basics Interview Questions

Q1. What is a Server Operating System?

Answer:

A Server Operating System is a specialized OS designed to provide services, manage users, control resources, and support multiple client machines in a network environment.

Unlike Windows 10/11 (client OS), Windows Server is built for:

-  Centralized management
-  Security
-  Networking
-  Multi-user environment
-  Enterprise services

Simple Explanation:

 Client machine uses services

 Server machine provides services

Example:

Machine	Role
Windows 11 Laptop 	Client
Windows Server 	Server

? Q2. What is the difference between Client OS and Server OS?

✓ Answer:

Client OS 

Server OS 

Used by end users Used to provide services

Windows 10/11 Windows Server

Limited connections Supports many users

Basic management Enterprise management

Personal use Company/business use

 Interview Tip:

Always mention:

👉 “Server OS is optimized for centralized management and enterprise-level services.”

? Q3. What are the common types of servers?

✓ Answer:

Server Type	Purpose
 Web Server	Hosts websites
 File Server	Stores files
 Mail Server	Email services
 DNS Server	Domain name resolution
 DHCP Server	IP assignment
 Active Directory Server	User authentication
 VPN Server	Secure remote access

Server Type

Purpose



VMware Server

Virtualization



Q4. Why do companies use Windows Server?



Answer:

Companies use Windows Server because managing hundreds or thousands of systems manually is impossible.

Windows Server provides:



Centralized control



User management



Security policies



Domain authentication



File sharing



Software deployment



Network management



Real-Life Example:

In a company with 1000 employees:

Without server:



Manual management

With server:



Centralized management



SECTION 2 – VMware & Server Installation Questions



Q5. Why do we install Windows Server in VMware?



Answer:

VMware is used to create virtual machines for:

- ✓ Testing
 - ✓ Learning
 - ✓ Lab practice
 - ✓ Server simulation
 - ✓ Safe experimentation
-

Advantages:

- No need for physical server
 - Snapshot support
 - Easy recovery
 - Multiple OS on one machine
-

Q6. Why should we NOT attach ISO immediately during VM creation?

✓ **Answer:**

Sometimes attaching ISO early may cause:

- ✗ Setup freeze
- ✗ Detection problems
- ✗ License page hang

So best practice is:

- ✓ Create VM hardware first
 - ✓ Attach ISO later
-

Interview-Level Explanation:

This is considered a virtualization troubleshooting technique.

? Q7. What hardware configuration is recommended for Windows Server VM?

✓ Answer:

Hardware Recommended

RAM  4 GB or more

CPU  2–4 cores

Disk  60 GB+

? Q8. Why do we select “Desktop Experience” edition?

✓ Answer:

Desktop Experience includes:

✓ GUI (Graphical User Interface)

Without Desktop Experience:

✗ Only command line available

 **Beginner-Friendly Explanation:**

GUI makes server management easier for IT support engineers.

? Q9. Installation stuck at “Getting files ready.” What could be the reason?

✓ Answer:

Possible causes:

Cause	Explanation
 Disk errors	HDD corruption
 Corrupted ISO	Bad setup files

Cause	Explanation
 Low storage	No extraction space
 Bad sectors	Disk damage

Troubleshooting:

Run:

```
chkdsk /f /r
```

 **Q10. What does CHKDSK do?**

 **Answer:**

CHKDSK checks disk integrity and repairs disk errors.

Important Switches:

Command Purpose

chkdsk /f Fix file system errors

chkdsk /r Recover bad sectors

SECTION 3 – Active Directory (AD) Questions

 **Q11. What is Active Directory?**

 **Answer:**

Active Directory (AD) is a Microsoft directory service used to centrally manage:

 Users 

 Computers 

 Policies 

✓ Printers 

✓ Resources 

 **Simple Explanation:**

 Company database + authentication system

 **Q12. What is a Domain Controller?**

✓ **Answer:**

A Domain Controller is a server where Active Directory Domain Services (AD DS) is installed.

It handles:

- ✓ Authentication
 - ✓ Authorization
 - ✓ Policy management
-

 **Q13. What is a Forest in Active Directory?**

✓ **Answer:**

A Forest is the highest-level structure in Active Directory that contains multiple domains.

 **Example:**

Microsoft may have:

- microsoft.com
- xbox.com
- office.com

Together they form:

 Forest

 Q14. Why is DNS important in Active Directory?

 Answer:

DNS helps clients locate the Domain Controller.

Without proper DNS:

-  Domain join fails
-  User login fails
-  GPO fails

 VERY IMPORTANT INTERVIEW POINT:

Always say:

 “Active Directory heavily depends on DNS.”

 Q15. What is OU (Organizational Unit)?

 Answer:

OU is a container inside Active Directory used to organize:

-  Users
-  Computers
-  Departments

 Example:

- HR OU
 - IT OU
 - Finance OU
-

? Q16. What is Group Policy (GPO)?

✓ Answer:

Group Policy is used to apply settings and restrictions centrally across domain systems.

🎯 Examples:

- ✓ Disable Control Panel
 - ✓ Block USB
 - ✓ Set wallpaper
 - ✓ Password policies
-

? Q17. What is the purpose of Group Policy?

✓ Answer:

Group Policy helps in:

- ✓ Security enforcement
 - ✓ User restrictions
 - ✓ Standardization
 - ✓ Automation
-

? Q18. What is DSRM Password?

✓ Answer:

DSRM stands for:

👉 Directory Services Restore Mode

It is used to recover Active Directory during disaster recovery situations.

Real Scenario:

If:

-  AD corrupted
-  Server crashed

Then:

-  DSRM helps recover domain services

SECTION 4 – Domain Joining Questions

Q19. What are the requirements for domain joining?

Answer:

Requirements:

-  Same network
-  Proper DNS
-  Reachable Domain Controller
-  Valid domain credentials

Q20. Why must DNS point to Server IP during domain joining?

Answer:

Because clients use DNS to locate the Domain Controller.

If DNS is wrong:

-  Domain join fails

Q21. How do you check network connectivity before domain joining?

Answer:

Use:

ping <IP>

Example:

ping 192.168.1.10

? Q22. How do you verify successful domain join?

✓ Answer:

Open:

sysdm.cpl

Check:

✓ Member of Domain: company.local

? Q23. What happens after user logs into domain first time?

✓ Answer:

After first login:

- ✓ User profile created**
 - ✓ GPO applied**
 - ✓ Policies activated**
 - ✓ Password change enforced**
-

SECTION 5 – Troubleshooting Questions

? Q24. Domain join failed. How will you troubleshoot?

✓ Professional Answer:

Step-by-Step Troubleshooting:

- 1 Check network connectivity**
- 2 Verify ping to server**
- 3 Check DNS settings**
- 4 Ensure server services running**

- 5 Verify username/password
 - 6 Check firewall
 - 7 Confirm same subnet/network
-

? Q25. GPO is not applying. What will you do?

✓ Answer:

Troubleshooting Steps:

- 1 Verify client joined to domain
 - 2 Check OU placement
 - 3 Verify GPO linked properly
 - 4 Run:
gpupdate /force
 - 5 Run:
gpresult /r
 - 6 Restart system
-

? Q26. Why can't client ping server?

✓ Possible Reasons:

Cause	Solution
Firewall blocking 🔥	Allow ICMP
Wrong IP 🌐	Reconfigure
Different network	Use same NAT/Bridge
Network adapter disabled	Enable adapter

? Q27. Active Directory Users & Computers not opening. What could be wrong?

✓ Answer:

Possible reasons:

- ✗ AD DS service stopped
- ✗ Corrupted RSAT tools
- ✗ DNS issue
- ✗ Permission issue

? Q28. Server running very slow. What will you check?

✓ Answer:

Check:

- ✓ RAM usage
- ✓ CPU utilization
- ✓ Disk usage
- ✓ Background services
- ✓ Event Viewer logs
- ✓ VMware resources

🔒 SECTION 6 – Security & Admin Questions

? Q29. Why is admin account important in IT support?

✓ Answer:

Admin account provides elevated privileges required for:

- ✓ Troubleshooting
- ✓ Installing software
- ✓ Changing settings
- ✓ Managing policies

? Q30. Why should normal users not have admin rights?

✓ Answer:

To prevent:

- ✗ Malware installation
- ✗ Unauthorized changes
- ✗ Security risks

? Q31. Why are Group Policies important for security?

✓ Answer:

GPO helps enforce:

- ✓ Password policies
- ✓ Device restrictions
- ✓ Security configurations
- ✓ User limitations



SECTION 7 – REAL INTERVIEW SCENARIO QUESTIONS

? Q32. User says Control Panel is blocked. What will you do?

✓ Answer:

- 1 Check if GPO applied
- 2 Login using admin account
- 3 Verify restriction policy
- 4 Modify GPO if needed
- 5 Run:

gpupdate /force

? Q33. Client machine cannot login using domain account.

✓ Answer:

Troubleshooting:

- ✓ Check internet/network
- ✓ Verify DNS
- ✓ Check AD account status
- ✓ Reset password
- ✓ Check time sync
- ✓ Ensure domain reachable

? Q34. Why is DNS called backbone of Active Directory?

✓ Answer:

Because every AD communication depends on DNS.

Without DNS:

- ✗ Users cannot login
 - ✗ Domain join fails
 - ✗ Policies fail
-

QUICK COMMANDS REVISION

Command	Purpose
ipconfig	Show IP
ping	Test connectivity
gpupdate /force	Apply GPO
gpresult /r	Check policies
chkdsk /f /r	Disk repair
sysdm.cpl	System/domain info

FINAL SUMMARY

Concept	Meaning
Windows Server 	Enterprise OS
AD 	Centralized management
Domain Controller 	Authentication server
OU 	Department grouping
GPO 	Policy management
DNS 	Domain resolution
VMware 	Virtualization
Domain Join 	Connect client to AD
